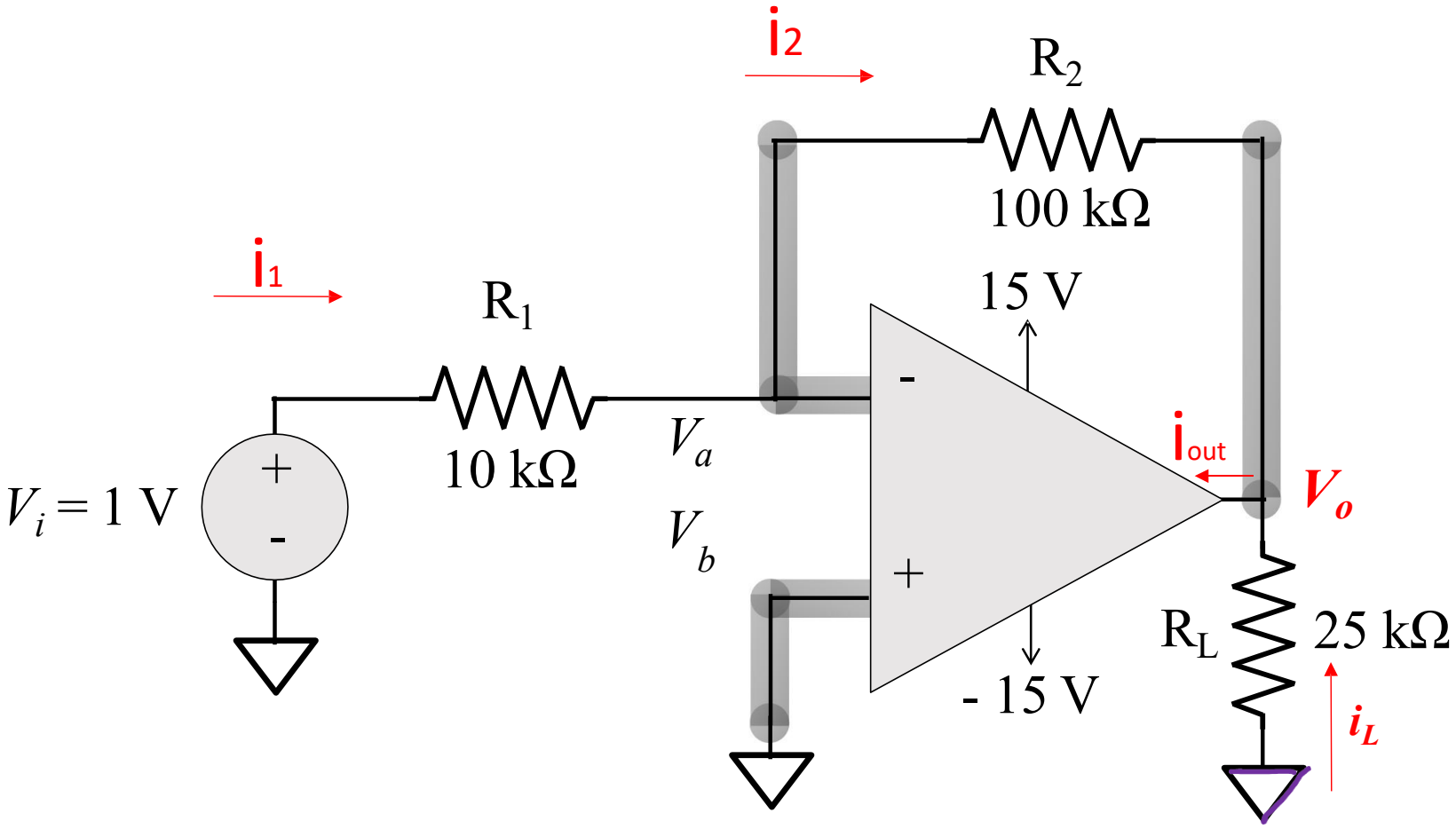
Analog Electronics II: Op-amp & 555 Timer

Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 5

Problem 10. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions. Calculate the A_{CL}, i_1, i_2, V_o, i_L, and i_{out}.



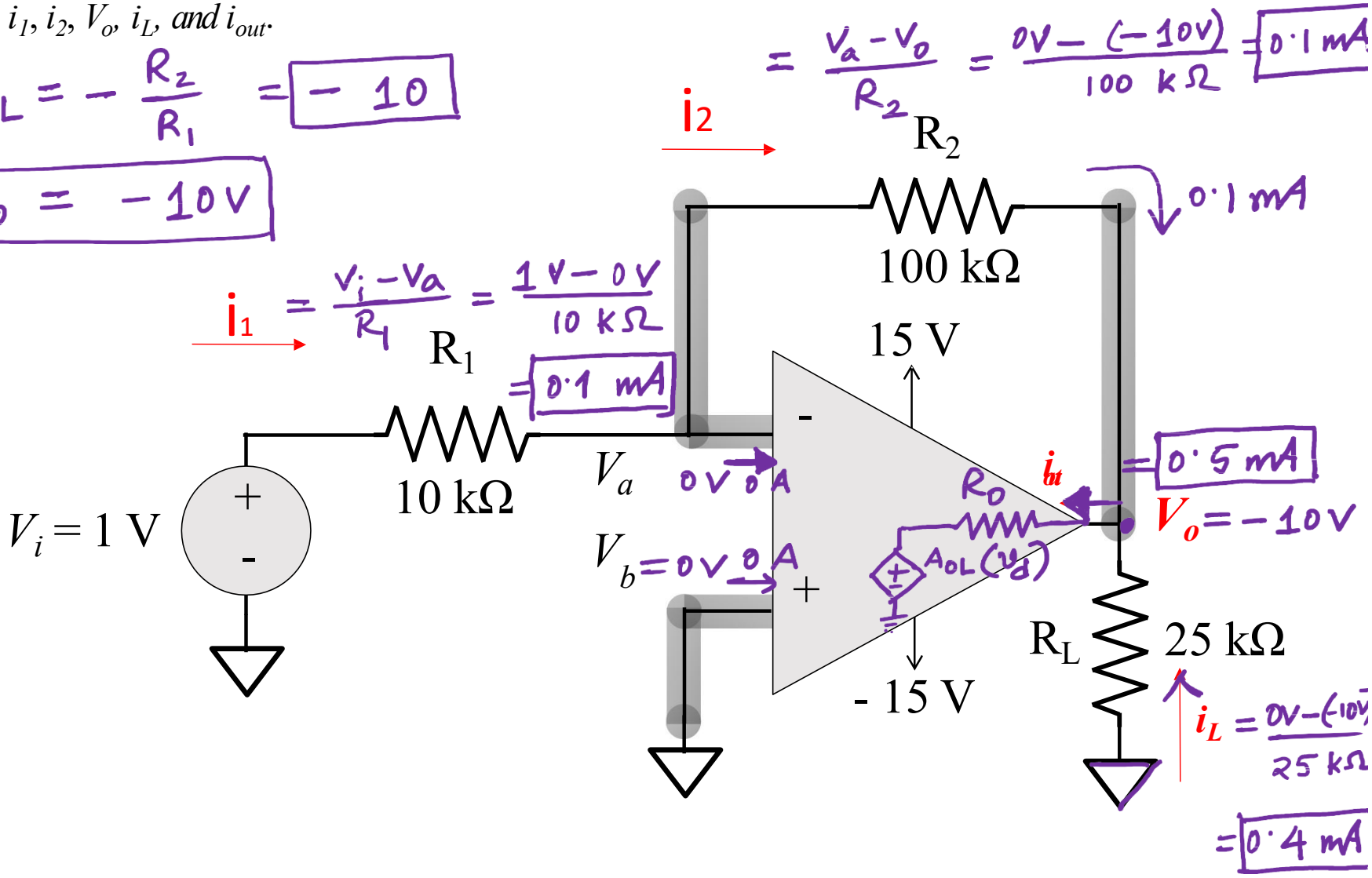
Problem 10. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions. Calculate the A_{CL}, i_1, i_2, V_o, i_L, and i_{out}.

$A_{CL} = -\frac{R_2}{R_1} = -10$

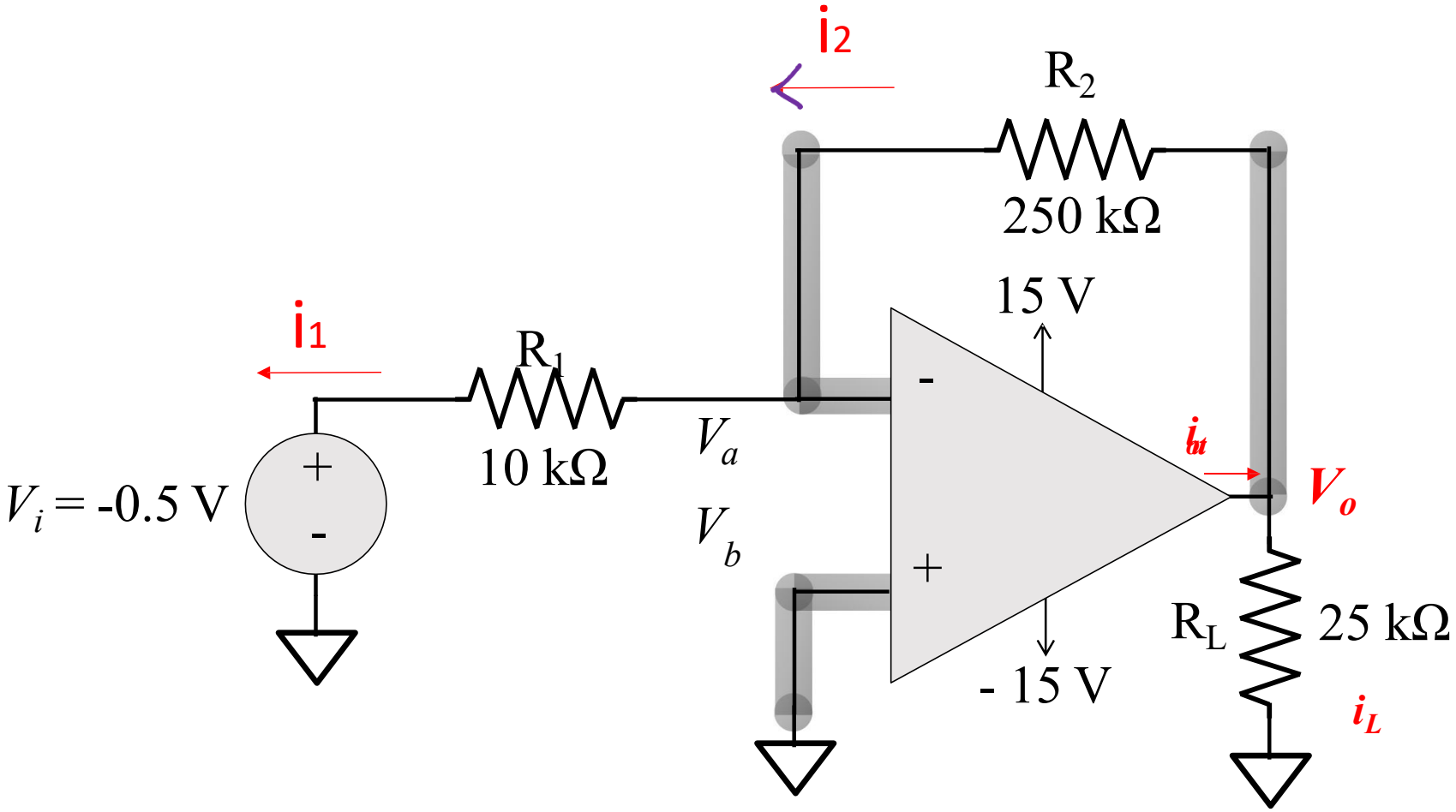
$V_o = -10$ V

$i_2 = \frac{V_a - V_o}{R_2} = \frac{0\text{V} - (-10\text{V})}{100\ \text{k}\Omega} = 0.1$ mA

$i_1 = \frac{V_i - V_a}{R_1} = \frac{1\text{V} - 0\text{V}}{10\ \text{k}\Omega} = 0.1$ mA



Problem 11. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions. Calculate the A_{CL}, i_1, i_2, V_o, i_L, and i_{out}.



Problem 11. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions. Calculate the A_{CL}, i_1, i_2, V_o, i_L, and i_{out}.

$A_{CL} = -25$

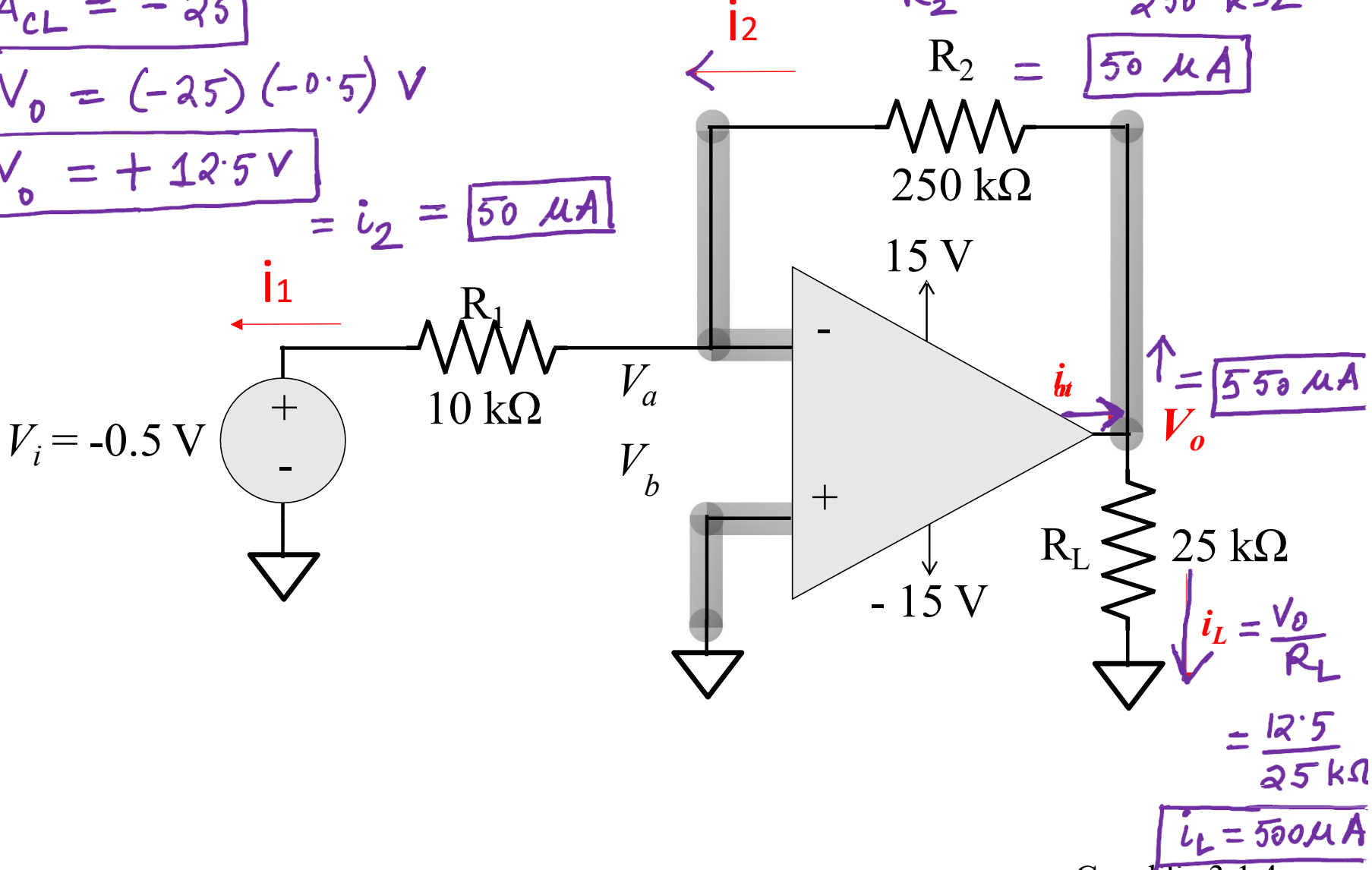
$V_o = (-25)(-0.5)$ V

$V_o = +12.5$ V

$= i_2 = 50\ \mu A$

$= \frac{V_o - V_a}{R_2} = \frac{12.5\text{ V} - 0\text{ V}}{250\ k\Omega}$

$R_2 = 50\ \mu A$



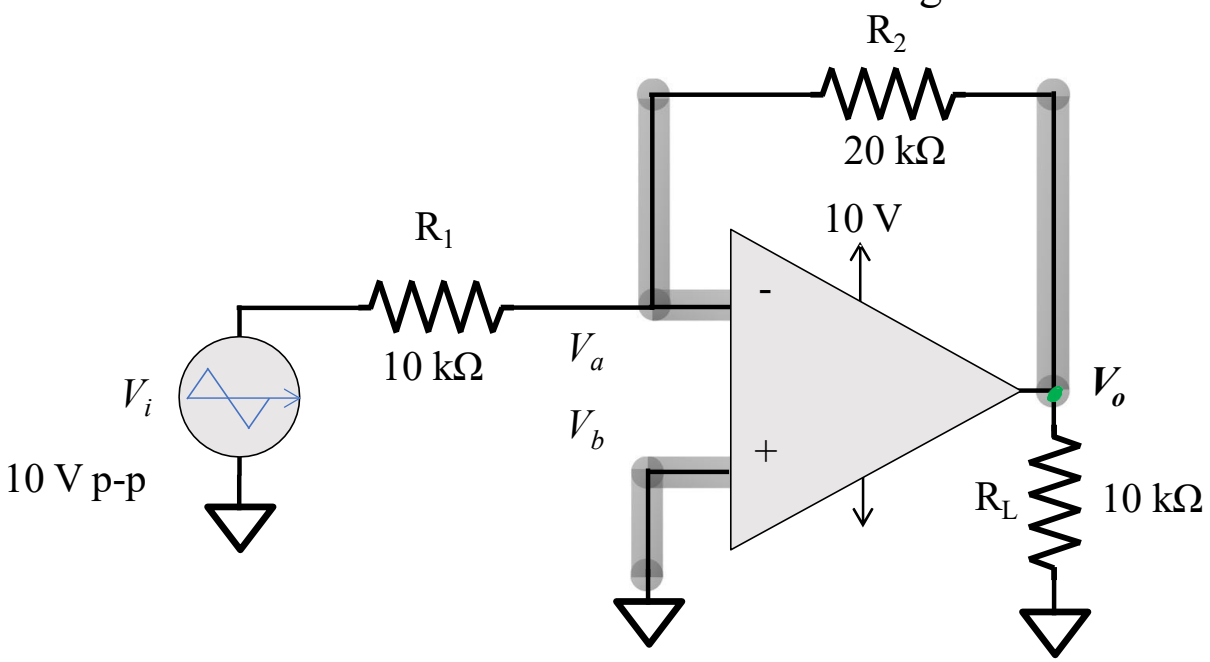
$= \frac{12.5}{25\ k\Omega}$

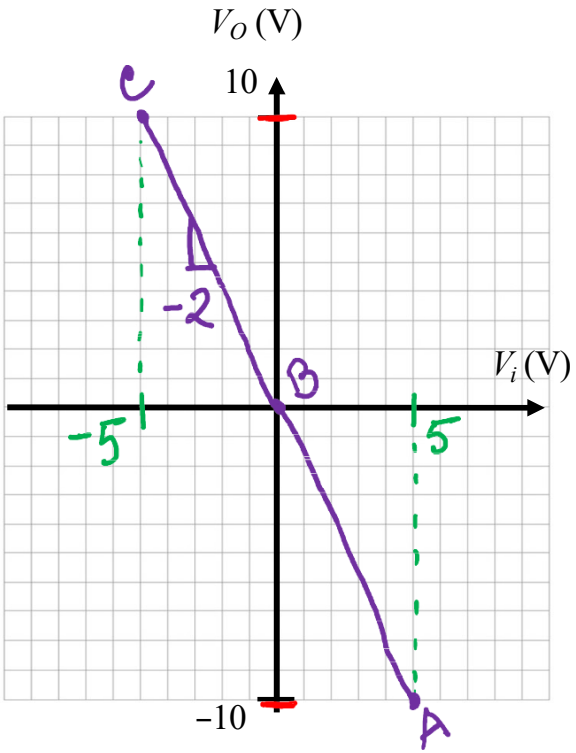
$i_L = 500\mu A$

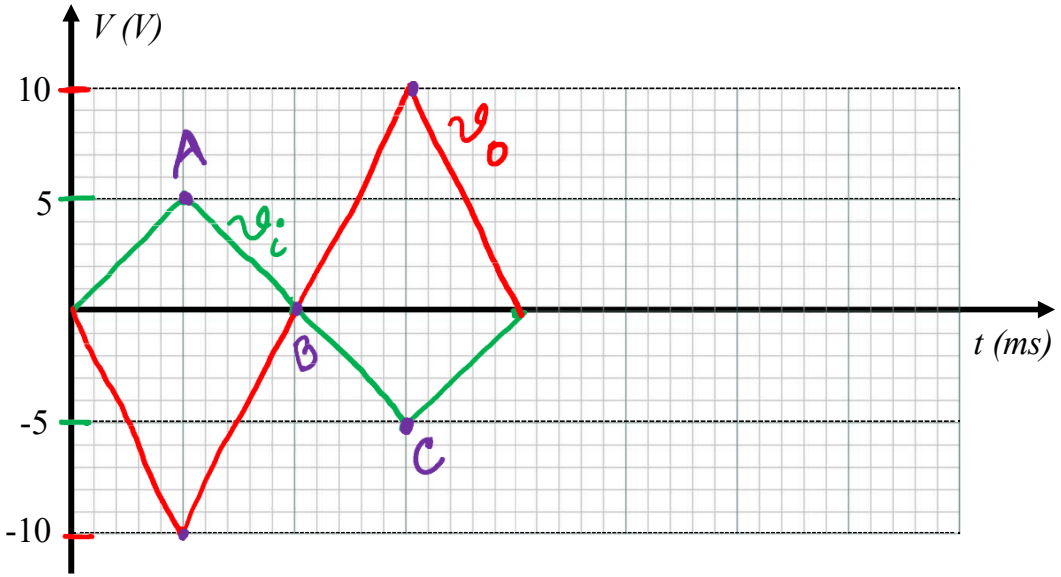
Problem 12. $A_{OL} = 2\times10^5$. $V_{sat+} = +10$ V, $V_{sat-} = -10$ V. Assume ideal conditions. Calculate the A_{CL}. Draw the V_i and V_o. Draw the transfer characteristics.

Coughlin 3-1.5

$A_{CL} = -2$

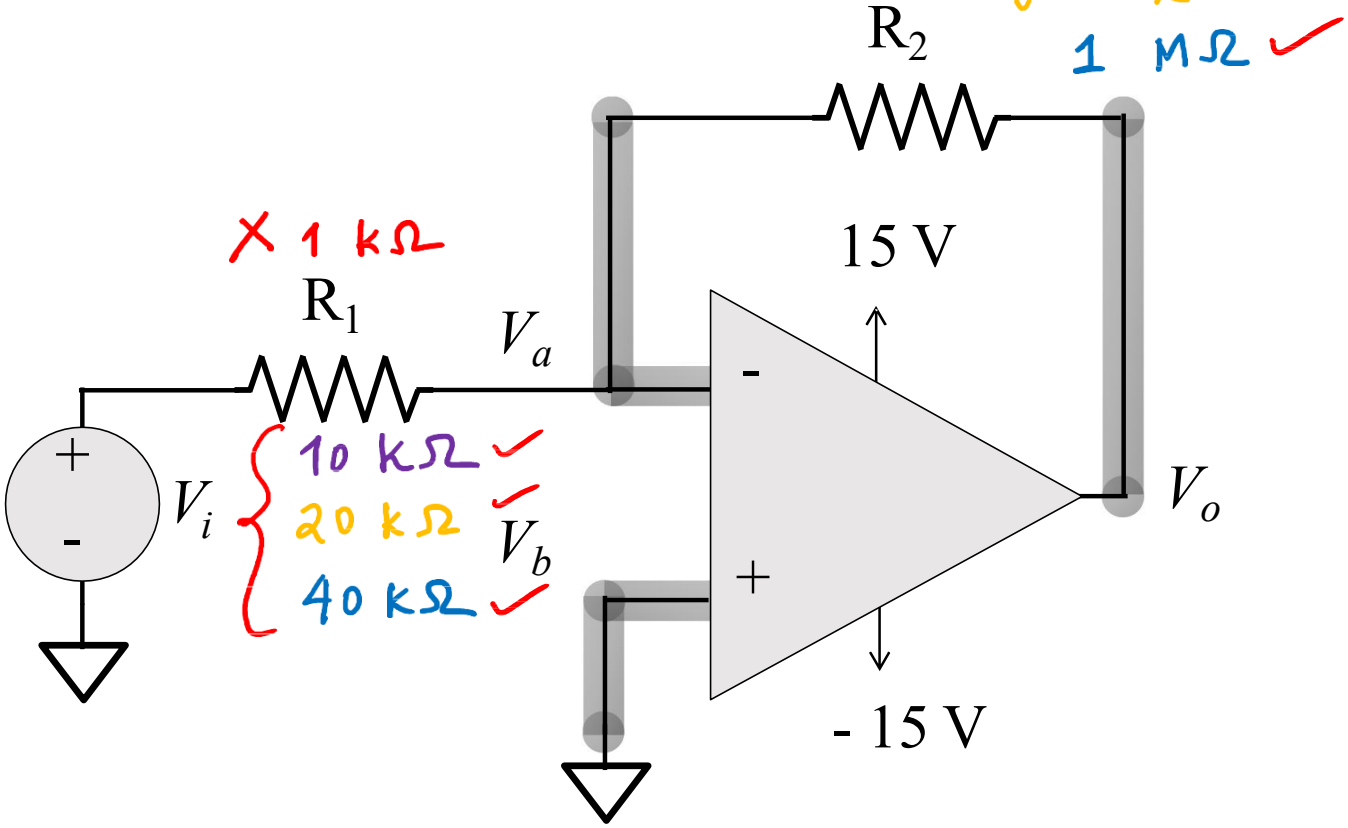




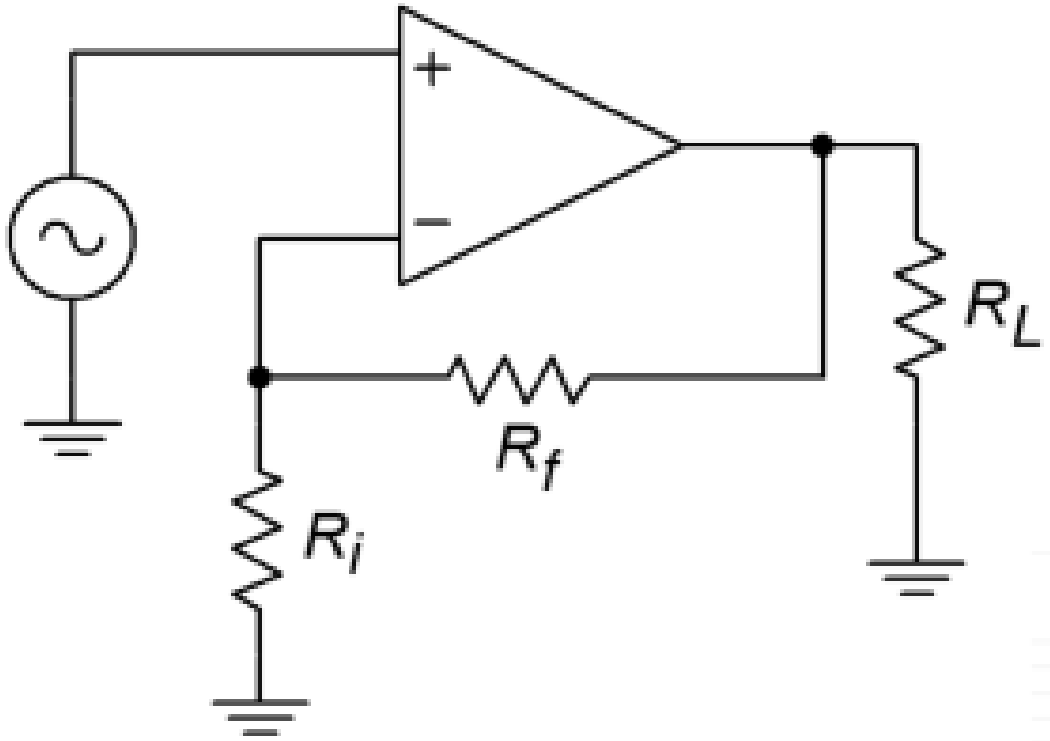


Problem 13. Design an amplifier with a gain of -25. The input resistance should be at least 10 kΩ. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions.

25 kΩ
250 kΩ ✓
500 kΩ ✓
1 MΩ ✓



The Noninverting Voltage Amplifier



$$A_v = \frac{V_{out}}{V_{in}}$$

$$V_{in} = V_{Ri}$$

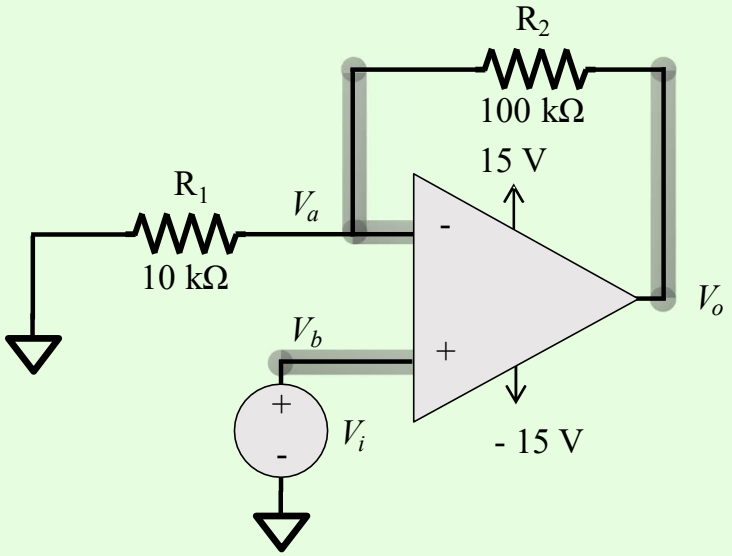
$$V_{out} = V_{Ri} + V_{Rf}$$

$$A_v = \frac{V_{Ri} + V_{Rf}}{V_{Ri}}$$

$$A_v = \frac{R_i I_{Ri} + R_f I_{Rf}}{R_i I_{Ri}} = \frac{R_i + R_f}{R_i} = 1 + \frac{R_f}{R_i}$$

Non-inverting amplifier gain $A_{CL} = 1 + \frac{R_2}{R_1}$

(a)

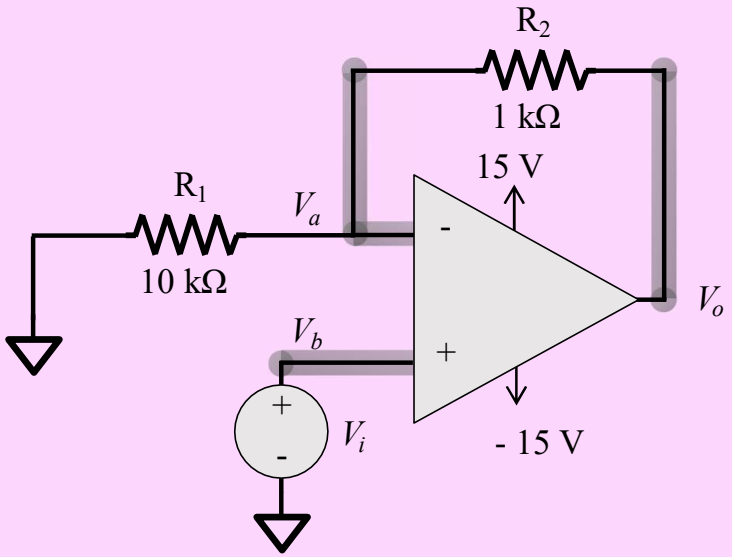


$A_{CL} = 1 + 10$

$= 11 \frac{V}{V}$

$A_{CL} > 1$: amplific^n

(b)



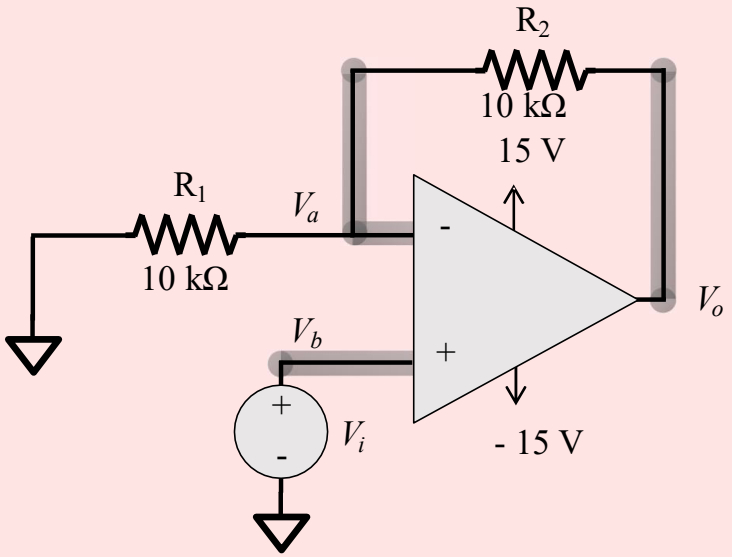
$A_{CL} = 1 + \frac{1}{10}$

$= 1.1 \frac{V}{V}$

$A_{CL} > 1$: amplific^n

Non-inverting amplifier gain A_{CL}

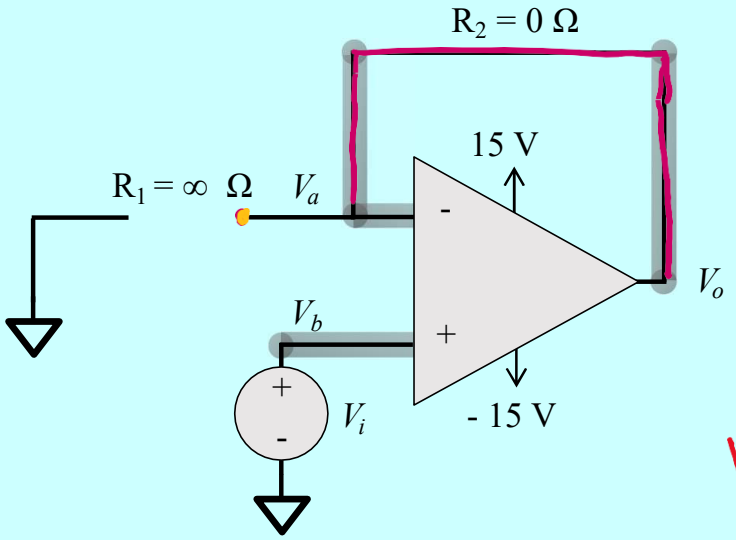
(c)



$A_{CL} = 1 + 1 = 2$

$A_{CL} > 1$: amplific^{n}

(d)

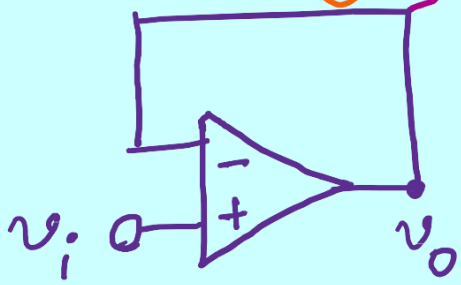


$A_{CL} = 1 + \frac{R_2}{R_1}$

$R_2 = 0\,\Omega$ (S.C.)

$R_1 = \infty\,\Omega$ (O.C.)

Both



Voltage follower / Buffer

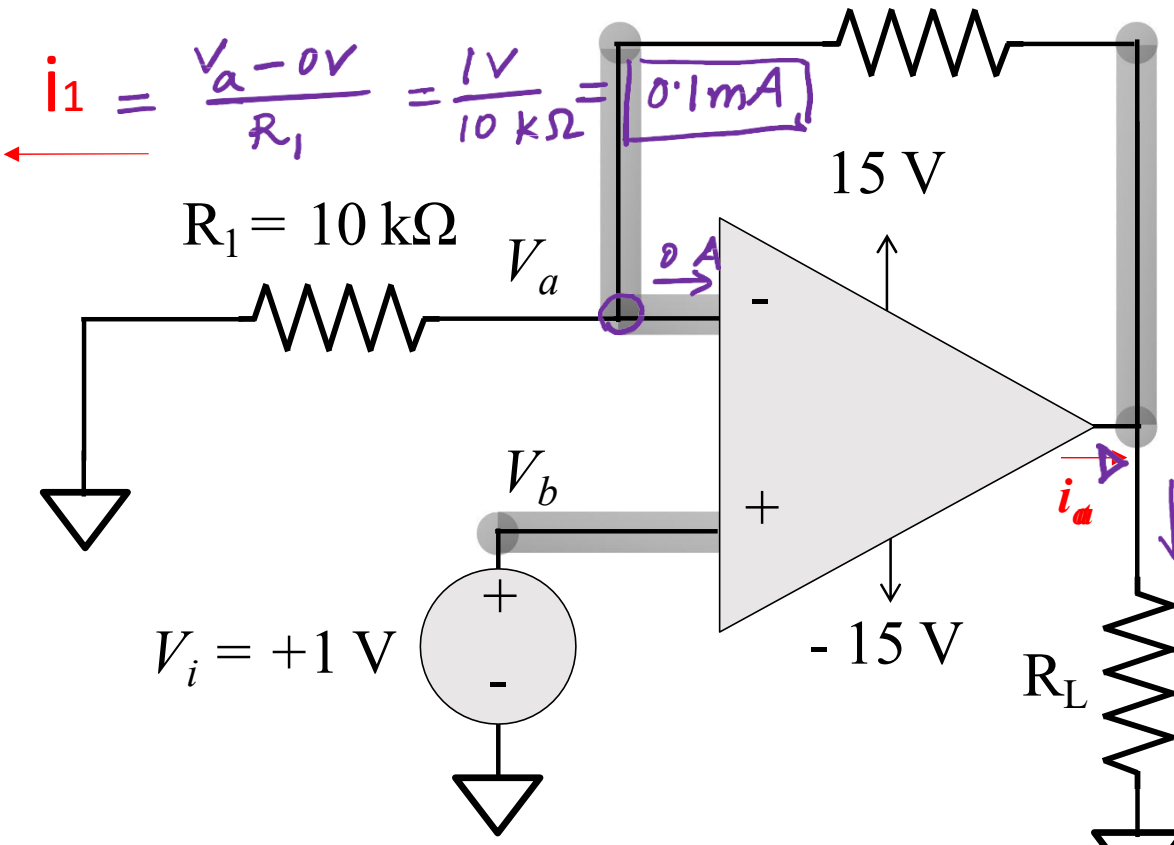
Problem 14. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions. Calculate the A_{CL}, i_1, i_2, V_o, i_L, and i_{out}.

$A_{CL} = 1 + \frac{R_2}{R_1} = 5$

$i_2 = i_1 = 0.1\text{ mA} = \frac{V_o - V_a}{R_2}$

$R_2 = 40\text{ k}\Omega = \frac{5\text{V} - 1\text{V}}{40\text{ k}\Omega} = \frac{4\text{V}}{40\text{ k}\Omega} = 0.1\text{ mA}$

$i_1 = \frac{V_a - 0\text{V}}{R_1} = \frac{1\text{V}}{10\text{ k}\Omega} = 0.1\text{mA}$



$V_o = +5\text{ V}$

$= i_2 + i_L$

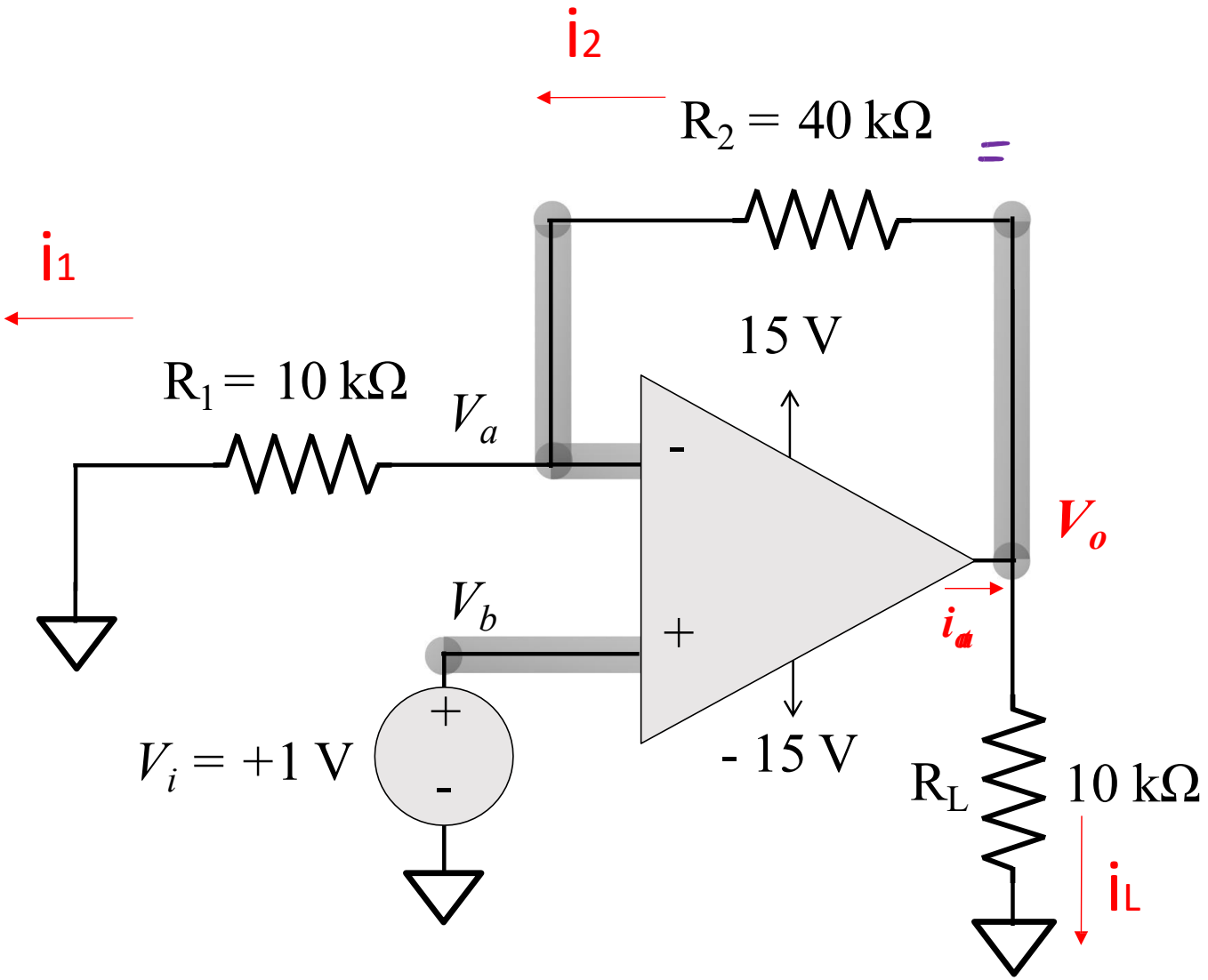
$= 0.6\text{ mA}$

R_L 10 kΩ

$i_L = \frac{V_o}{R_L} = \frac{5\text{V}}{10\text{ k}\Omega}$

$= 0.5\text{ mA}$

Problem 14. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions. Calculate the A_{CL}, i_1, i_2, V_o, i_L, and i_{out}.

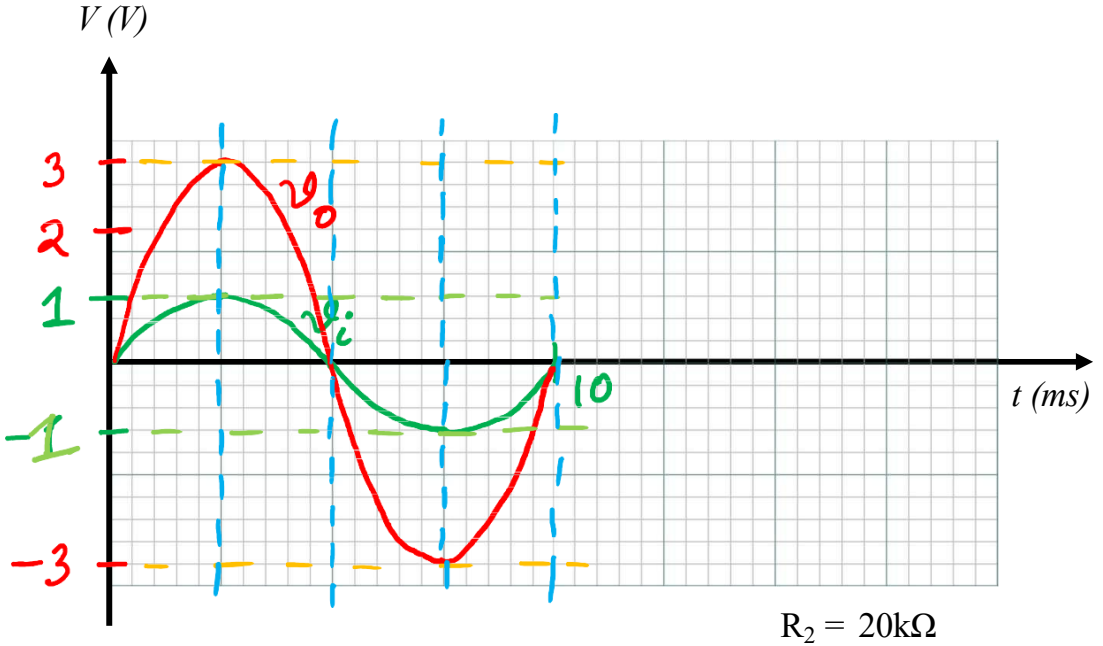


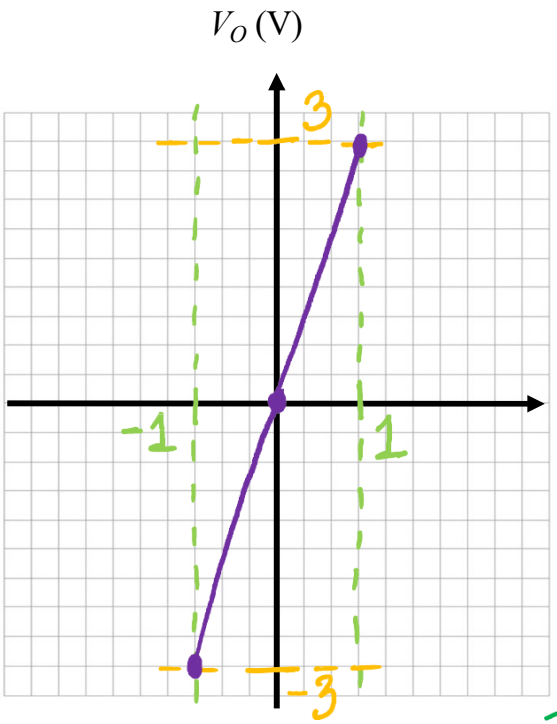
Problem 15. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions. Calculate the A_{CL}. Draw the V_i and V_o. Draw the transfer characteristics.

$A_{CL} = 1 + \frac{R_2}{R_1} = 1 + 2 = 3$

Non-inv

+ve slope



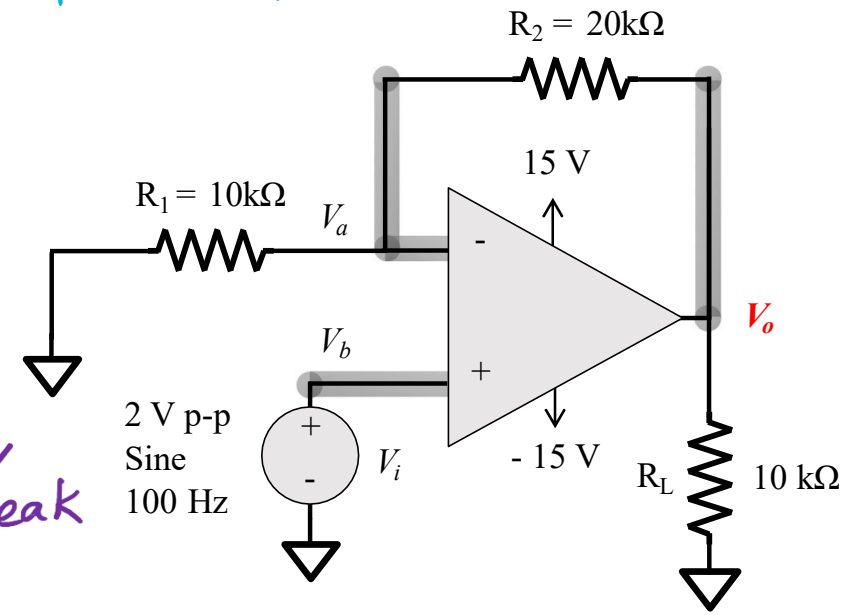


$T = \frac{1}{10}$ s

$= 0.01$ s

10 ms

1V peak



Problem 16. Design an amplifier with a gain of +25. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions.

Coughlin Example 3-14

Practice Problem: Design a three-stage amplifier with gain +10, -18 and -27. Use a 270kΩ feedback resistor for all three stages. What will be the output voltage if input voltage is 150μV?

For the gain of +10,

$$A_1 = 1 + \frac{R_f}{R_1} = +10$$

$$\frac{R_f}{R_1} = 10 - 1 = 9$$

$$R_1 = \frac{R_f}{9} = \frac{270\,\text{k}\Omega}{9} = 30\,\text{k}\Omega$$

For the gain of −18,

$$A_2 = -\frac{R_f}{R_2} = -18$$

$$R_2 = \frac{R_f}{18} = \frac{270\,\text{k}\Omega}{18} = 15\,\text{k}\Omega$$

For the gain of −27,

$$A_3 = -\frac{R_f}{R_3} = -27$$

$$R_3 = \frac{R_f}{27} = \frac{270\,\text{k}\Omega}{27} = 10\,\text{k}\Omega$$

$$V_o = A_1A_2A_3V_1 = (10)(-18)(-27)(150\ \mu\text{V}) = 4860(150\ \mu\text{V})$$
$$= \mathbf{0.729\ V}$$

